TRIGONOMETRIC FUNCTIONS

DIFFERENTIAL CALCULUS

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

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SEATWORK

0

ABSENT

3

45-min LATE

5 max

15-min LATE

TOPIC OUTLINE

Trigonometric Identities

Derivatives of Trigonometric Functions

TRIGONOMETRIC IDENTITIES

RECIPROCAL AND RATIO

$$\csc\theta = \frac{1}{\sin\theta}$$

$$\sec\theta = \frac{1}{\cos\theta}$$

$$\cot\theta = \frac{1}{\tan\theta}$$

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

$$\cot\theta = \frac{\cos\theta}{\sin\theta}$$

PYTHAGOREAN

$$\cos^2\theta + \sin^2\theta = 1$$

$$1 + \tan^2\theta = \sec^2\theta$$

$$1 + \cot^2\theta = \csc^2\theta$$

DERIVATIVES OF TRIGONOMETRIC FUNCTIONS

$$\frac{d}{dx}(\sin x) = \cos x$$

Differentiate $y = x^2 \sin x$

$$\frac{dy}{dx} = x^2 \frac{d}{dx}(\sin x) + \sin x \frac{d}{dx}(x^2)$$

$$\frac{dy}{dx} = x^2 \cos x + \sin x\,(2x)$$

$$\boxed{\frac{dy}{dx} = x^2 \cos x + 2x \sin x}$$

ans

$$\frac{d}{dx}(\cos x) = -\sin x$$

Differentiate $f(x) = 3x^2 - 2\cos x$

$$\frac{df(x)}{dx} = \frac{d}{dx}(3x^2) - 2\frac{d}{dx}(\cos x)$$

$$\frac{df(x)}{dx} = 6x - 2(-\sin x)$$

$$\boxed{\frac{df(x)}{dx} = 6x + 2\sin x}$$

ans

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

Differentiate $y = \tan x$

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right)$$

$$\frac{dy}{dx} = \frac{\cos x \frac{d}{dx}(\sin x) - \sin x \frac{d}{dx}(\cos x)}{(\cos x)^2}$$

$$\frac{dy}{dx} = \frac{\cos x\,(\cos x) - \sin x(-\sin x)}{(\cos x)^2}$$

$$\frac{dy}{dx} = \frac{\cos^2 x + \sin^2 x}{(\cos x)^2}$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{dy}{dx} = \frac{1}{(\cos x)^2}$$

$$\frac{dy}{dx} = (\sec x)^2$$

$$\boxed{\frac{dy}{dx} = \sec^2 x}$$

ans

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

Differentiate $y = \sec x$

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{1}{\cos x}\right)$$

$$\frac{dy}{dx} = \frac{\cos x \frac{d}{dx}(1) - 1\frac{d}{dx}(\cos x)}{(\cos x)^2}$$

$$\frac{dy}{dx} = \frac{\cos x\,(0) - (-\sin x)}{(\cos x)^2}$$

$$\frac{dy}{dx} = \frac{\sin x}{(\cos x)^2}$$

$$\frac{dy}{dx} = \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x}$$

$$\boxed{\frac{dy}{dx} = \sec x \tan x}$$

ans

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

Differentiate $y = \csc x$

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{1}{\sin x}\right)$$

$$\frac{dy}{dx} = \frac{\sin x \frac{d}{dx}(1) - 1\frac{d}{dx}(\sin x)}{(\sin x)^2}$$

$$\frac{dy}{dx} = \frac{\sin x\,(0) - (\cos x)}{(\sin x)^2}$$

$$\frac{dy}{dx} = \frac{-\cos x}{(\sin x)^2}$$

$$\frac{dy}{dx} = \frac{1}{\sin x} \cdot \frac{-\cos x}{\sin x}$$

$$\boxed{\frac{dy}{dx} = -\csc x \cot x}$$

ans

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

Differentiate $y = \cot x$

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{\cos x}{\sin x}\right)$$

$$\frac{dy}{dx} = \frac{\sin x \frac{d}{dx}(\cos x) - \cos x \frac{d}{dx}(\sin x)}{(\sin x)^2}$$

$$\frac{dy}{dx} = \frac{\sin x\,(-\sin x) - \cos x\,(\cos x)}{(\sin x)^2}$$

$$\frac{dy}{dx} = \frac{-\sin^2 x - \cos^2 x}{(\sin x)^2}$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{dy}{dx} = \frac{-(\sin^2 x + \cos^2 x)}{(\sin x)^2}$$

$$\frac{dy}{dx} = \frac{-1}{(\sin x)^2}$$

$$\boxed{\frac{dy}{dx} = -\csc^2 x}$$

ans

Differentiate $y = \sqrt{x}\sin x$

$$\frac{dy}{dx} = \sin x \frac{d}{dx}\sqrt{x} + \sqrt{x}\frac{d}{dx}\sin x$$

$$\frac{dy}{dx} = \sin x \frac{d}{dx}x^{\frac{1}{2}} + \sqrt{x}\frac{d}{dx}\sin x$$

$$\frac{dy}{dx} = \sin x\left(\frac{1}{2}x^{-\frac{1}{2}}\right) + \sqrt{x}\cos x$$

$$\frac{dy}{dx} = \frac{\sin x}{2\sqrt{x}} + \sqrt{x}\cos x$$

$$\frac{dy}{dx} = \frac{\sin x}{2\sqrt{x}} \cdot \frac{\sqrt{x}}{\sqrt{x}} + \sqrt{x}\cos x$$

$$\frac{dy}{dx} = \frac{\sqrt{x}\sin x}{2x} + \sqrt{x}\cos x$$

$$\boxed{\frac{dy}{dx} = \sqrt{x}\left(\cos x + \frac{\sin x}{2x}\right)}$$

ans

Differentiate $u = \dfrac{\cos x}{1 - \sin x}$

$$\frac{du}{dx} = \frac{(1 - \sin x)\frac{d}{dx}(\cos x) - \cos x \frac{d}{dx}(1 - \sin x)}{(1 - \sin x)^2}$$

$$\frac{du}{dx} = \frac{(1 - \sin x)(-\sin x) - \cos x(-\cos x)}{(1 - \sin x)^2}$$

$$\frac{du}{dx} = \frac{-\sin x + \sin^2 x + \cos^2 x}{(1 - \sin x)^2}$$

$$\frac{du}{dx} = \frac{1 - \sin x}{(1 - \sin x)^2}$$

$$\boxed{\frac{du}{dx} = \frac{1}{1 - \sin x}}$$

ans

Differentiate $u = \dfrac{\sec\theta}{1 + \sec\theta}$

$$\frac{du}{d\theta} = \frac{(1+\sec\theta)\dfrac{d}{d\theta}(\sec\theta) - \sec\theta\dfrac{d}{d\theta}(1+\sec\theta)}{(1+\sec\theta)^2}$$

$$\frac{du}{d\theta} = \frac{(1+\sec\theta)(\sec\theta\tan\theta) - \sec\theta(\sec\theta\tan\theta)}{(1+\sec\theta)^2}$$

$$\frac{du}{d\theta} = \frac{\left(\sec\theta\tan\theta + \sec^2\theta\tan\theta\right) - \sec^2\theta\tan\theta}{(1+\sec\theta)^2}$$

$$\boxed{\frac{du}{d\theta} = \frac{\sec\theta\tan\theta}{(1+\sec\theta)^2}}$$

ans

SEATWORK